Predictors of Religiosity and Spirituality

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Abstract

The level of one's religiosity and spirituality are often associated with one's other beliefs and habits regarding issues considered related in nature to faith, including political beliefs, family relationships, church attendance and prayer habits, among others. After some preliminary exploratory data analysis to get a feel for the spread of our data, we analyzed the group differences between faith traditions, the relationship between spirituality and religiosity, behavioral and political belief predictors of religiosity and spirituality, finally culminating in the complete models that predict the spirituality and religiosity of individuals based on most available factors. The complete models were able to predict much more accurately the religiosity of an individual, with an R^2 of .71 for the religiosity model and an R^2 of .43 for the spirituality model.

Introduction

The dataset that we're working with comes from the PEW Research Center. The survey was intended to get responses from as many Americans as possible, offering surveys in English and Spanish in all 50 states, with online, paper and phone format options. The data contains information about the religious, political, spiritual and social background of those surveyed. The data can be accessed at the following link: https://www.pewresearch.org/dataset/2023-24-religious-landscape-study-rls-dataset/ After trimming down our dataset to include only those with useful information and those without all NA values, we were left with 93 columns. Our driving question is twofold: 1) How well can we predict religiosity and spirituality based on factors including church attendance, family relationships, marriage, political opinions, religious affiliation and devotion, etc.? 2) Which factors are most effective in predicting religiosity/spirituality of individuals? Does this vary by religious affiliation?

Exploratory Data Analysis

In our dataset, many variables were unusable due to lack of data or relevance to our research question. We narrowed our dataset down to 93 variables, of which the 28 most important are:

RELTRAD (religious affiliation)

CURREL (same as before but collapses evangelical, mainline and black prot into one group)

RELPER (how religious the respondent is, 1-4)

SPIRPER (how spiritual the respondent is, 1-4)

ATTNDPERRLS (how often they attend in person)

GOD (belief in God)

HVN (is there a heaven)

HLL (is there a hell)

PRAY (frequency of prayer in personal life)

PRAC_B (how often do you read scripture outside of worship service)

PRAC_D (how often do you share your beliefs with others)

ABRTLGL (support for legalization of abortion in all or most cases)

FRMREL (religion of family)

CHATTEND (how often you attended religious services growing up)

MARITAL (marital status)

SCPRY2 (prayers in public schools)

BIRTHDECADE (birth decade)

HISP (hispanic or not)

RACECMB (race)

EDUCREC (highest educational attainment)

E2 (currently enrolled in school)

EMPLSIT (current work situation)

USGEN (immigrant or child or immigrants)

REG (registered to vote or not)

PARTY (political party affiliation)

IDEO (political views)

INC_SDT1 (income bracket)

GENDER (male or female or other)

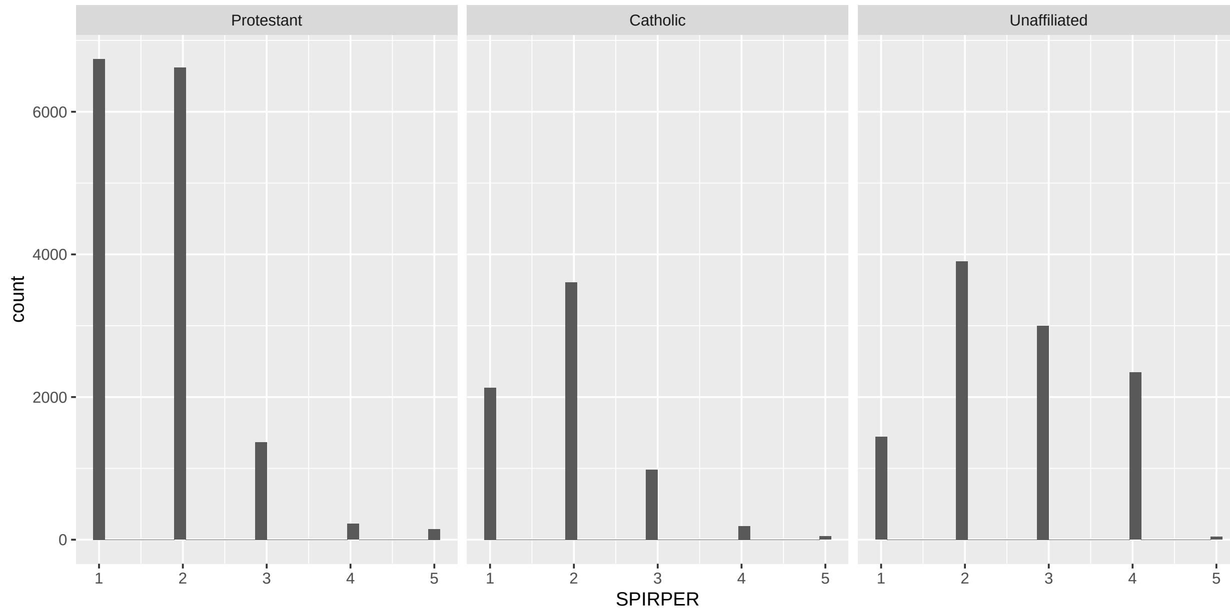
Many other variables exist and are used in this paper, but these are a sampling of many that are related to the question of religiosity and spirituality.

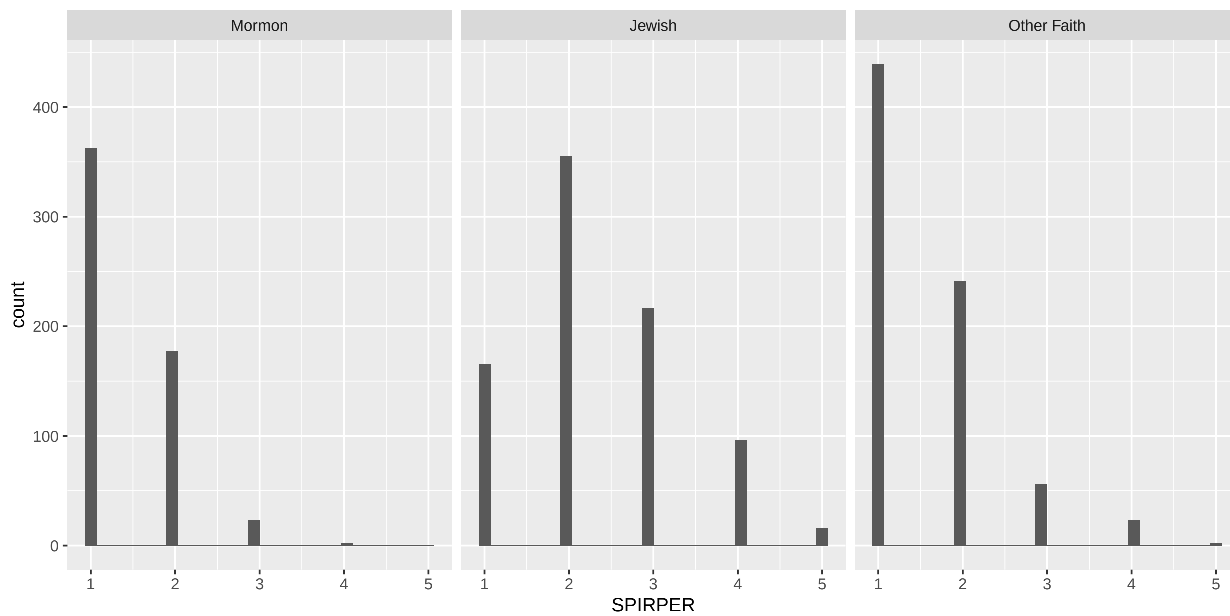
First, we turned CURREL into a factor so that we can create "bins" of how many people fall into each religious tradition.

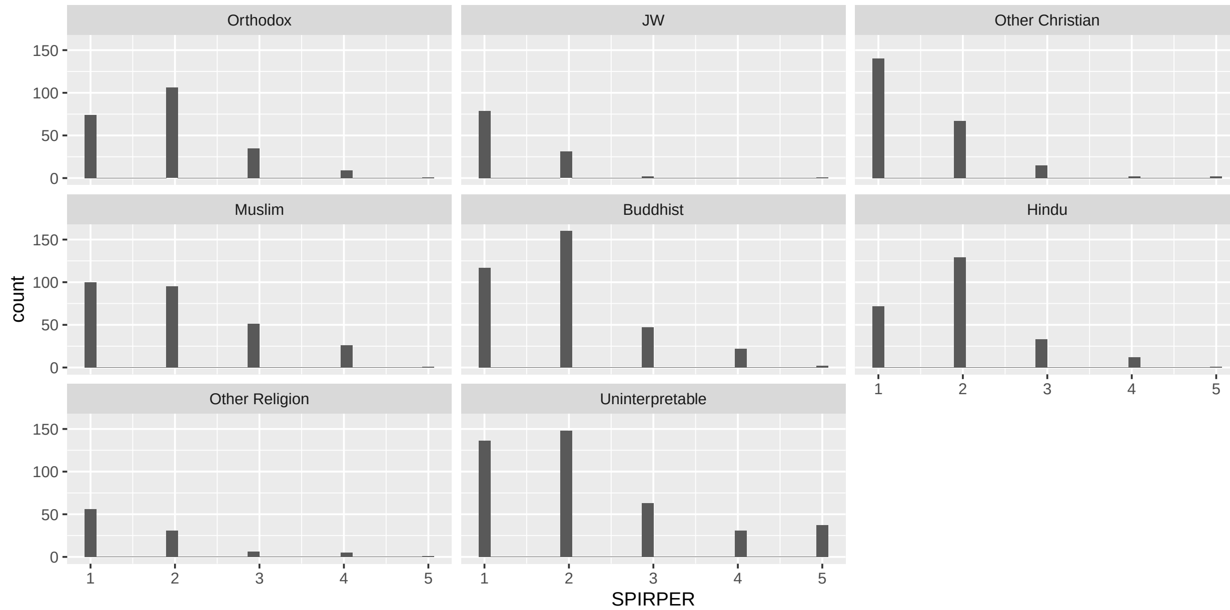
From there, we were able to look at the relationship between some of the other variables and religion.

First, before trying to plot the data, we noticed that there was a much larger number of those who identified as Catholic, Protestant or Unaffiliated than anything else, so we split the histogram facet_wrap into two to account for this (then split it again to account for the much larger prevalence of Mormon, Jewish and Other Faith than all the other religious identities combined) so that the spread of each religion across the different values would be clear. This splitting up of the religious categories will prove useful throughout the duration of the EDA section.

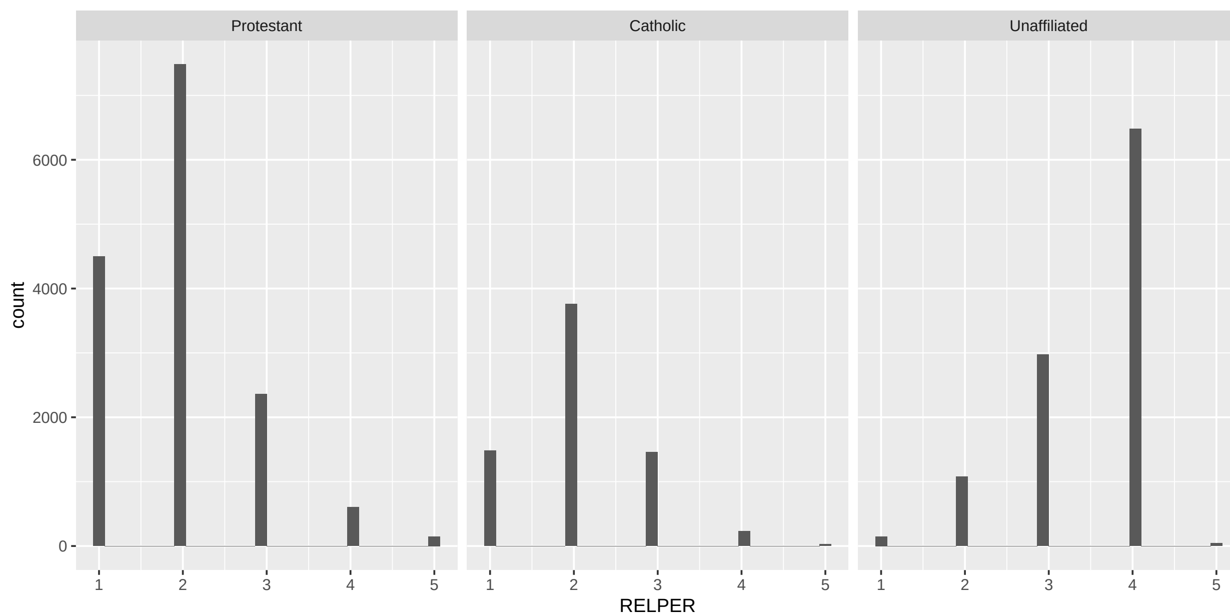
Looking at the correlation between each specific faith and personal spirituality (1=most spiritual, 4= least spiritual, 5=don't know/refused), we notice a few trends: most faiths seemed to have the largest spike at either 1 or 2. Interestingly, the Unaffiliated category also has the largest spikes around 1 and 2, and not 4. This could be that many people who identified as Unaffiliated, although they did not profess a specific religion, still considered themselves spiritual (spiritual but not religious).

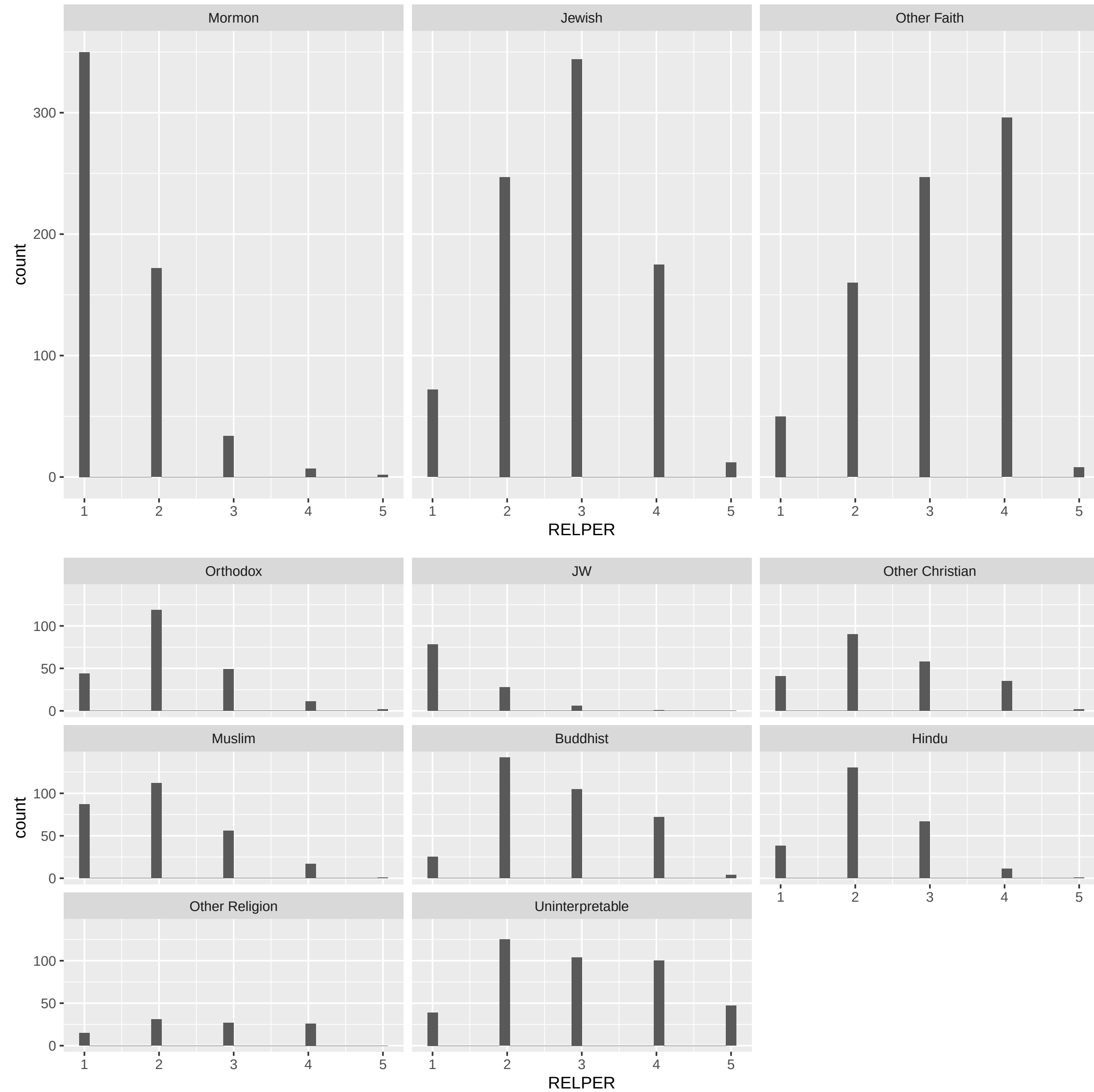

Protestant
Catholic
Unaffiliated
count
6000
4000
2000
0
1
2
3
4
5
SPIRPER


Mormon
Jewish
Other Faith
count
400
300
200
100
0
1
2
3
4
5
SPIRPER



The spread for religiosity is shifted more to the higher (less religious) values than spirituality: there were only two religions where the plurality (largest spike) was 1 (most religious).





Statistical Analysis

To formally investigate the relationships suggested by the exploratory data analysis, we apply several statistical methods, including analysis of variance (ANOVA), correlation analysis, simple linear regression, and multiple linear regression. These methods allow us to test for group differences, measure associations between variables, and identify key predictors of religiosity.

Section 1: Group Differences

We begin by examining whether religiosity differs across religious traditions.

Hypothesis Test: H_0: All religious groups have the same mean religiosity.

H_a: At least one group's mean is different.

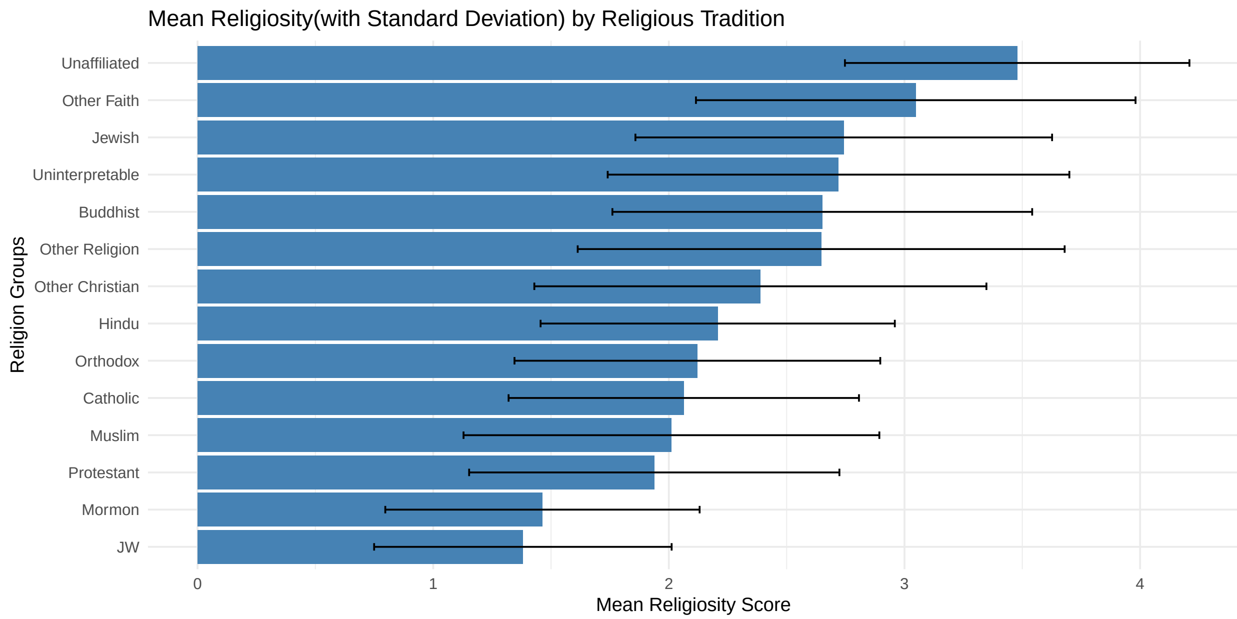
```
##                Df Sum Sq Mean Sq F value Pr(>F)
## CURREL         13  17394  1338.0    2247 <2e-16 ***
## Residuals   36590  21785     0.6
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## # A tibble: 14 x 4
##    CURREL          mean_religiosity sd_religiosity     n
##    <fct>                      <dbl>          <dbl> <int>
##  1 JW                          1.38          0.631   113
##  2 Mormon                      1.46          0.667   563
##  3 Protestant                  1.94          0.786 14949
##  4 Muslim                      2.01          0.882   272
##  5 Catholic                    2.06          0.743  6928
##  6 Orthodox                    2.12          0.776   223
##  7 Hindu                       2.21          0.752   246
##  8 Other Christian             2.39          0.959   224
##  9 Other Religion              2.65          1.03     99
## 10 Buddhist                    2.65          0.891   344
## 11 Uninterpretable             2.72          0.980   368
## 12 Jewish                      2.74          0.884   838
## 13 Other Faith                 3.05          0.933   753
## 14 Unaffiliated                3.48          0.731 10684
```

The ANOVA results indicate that we reject the null hypothesis, as the p-value is less than 0.05. This suggests that religiosity differs significantly across religious traditions.

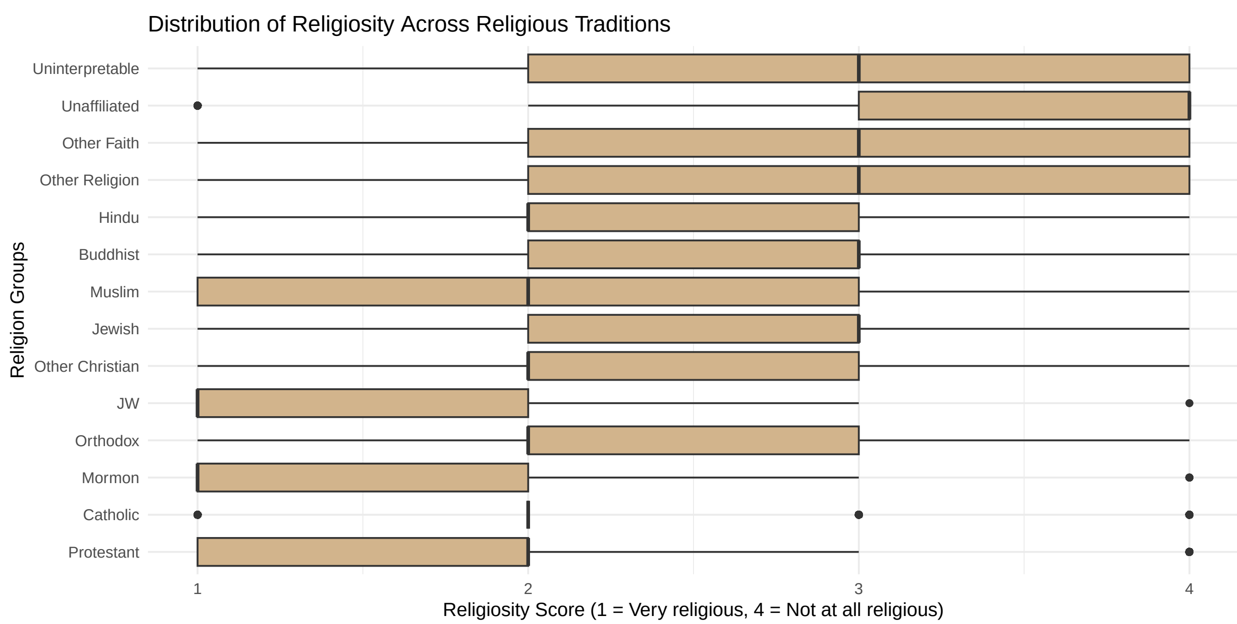
To further examine these differences, we computed the mean religiosity score for each group. Religiosity is measured on a scale from 1 (very religious) to 4 (not at all religious), excluding responses coded as 5 (don't know/refused). Because lower values indicate higher religiosity, groups with lower mean scores are more religious on average. The results show that Protestants have lower mean religiosity scores than Catholics and the unaffiliated, indicating higher average religiosity, while the unaffiliated group reports the highest mean scores, indicating the lowest levels of religiosity.

Protestants, Catholics, and the unaffiliated also have the largest sample sizes, making these estimates more reliable. The standard deviations, which range approximately from 0.5 to 1, suggest a moderate level of variation in religiosity within each group.



Check Assumptions for ANOVA: 1. Independence

2. Normality within groups
3. Common variance



Independence is assumed based on the sampling design of the dataset.

A boxplot of religiosity scores was used to examine the distribution and variability of religiosity across religious traditions. The plot shows the median, interquartile range, and potential outliers for each group.

The distributions appear reasonably symmetric within most groups, with no extreme deviations, suggesting that the assumption of approximate normality is reasonable. In addition, the spread of scores across groups is relatively similar, providing preliminary evidence that the assumption of equal variances is satisfied.

Section 2: Spirituality vs Religiosity

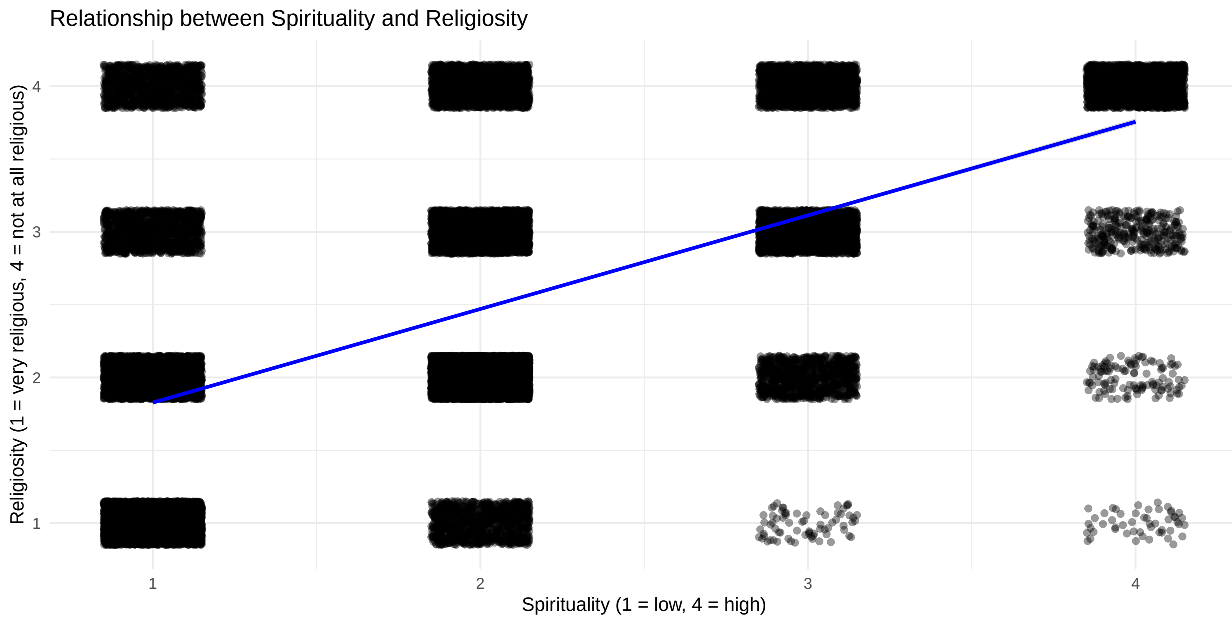
Next, we examine the relationship between spirituality and religiosity, as the exploratory data analysis suggested that even unaffiliated individuals report relatively high levels of spirituality.

```
## [1] 0.561914
```

```
##
## Call:
## lm(formula = RELPER ~ SPIRPER, data = data_no5)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.7564 -0.4704 -0.1134  0.5296  2.1726
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 1.184407   0.010869   109.0   <2e-16 ***
## SPIRPER     0.642997   0.004959   129.7   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8557 on 36437 degrees of freedom
## Multiple R-squared:  0.3157, Adjusted R-squared:  0.3157
## F-statistic: 1.681e+04 on 1 and 36437 DF,  p-value: < 2.2e-16
```

A correlation analysis was conducted between religiosity (RELPER) and spirituality (SPIRPER), excluding missing responses. The result is 0.562, indicating a moderate positive relationship between the two variables. This suggests that higher levels of spirituality are associated with higher religiosity scores. Given that higher religiosity scores correspond to lower levels of religiosity (1 = very religious, 4 = not at all religious), this indicates that higher spirituality is associated with lower reported religiosity.

A simple linear regression further examined this relationship. The results show that spirituality is a statistically significant predictor of religiosity ($p < 0.05$). The coefficient indicates that a one-unit increase in spirituality is associated with an increase in religiosity score. The model explains approximately 31.57% ($R^2 = 0.3157$) of the variation in religiosity, suggesting a moderate relationship.



As spirituality increases, religiosity score increased, indicating lower religiosity. A jittered scatterplot with a fitted linear regression line was used to visualize the relationship between spirituality and religiosity. Jittering was applied to help with data visualization due to the discrete nature of the 1–4 response scale.

Section 3: Behavioral Predictors

While spirituality captures internal beliefs, religiosity may also be influenced by observable behaviors such as attendance and prayer. We therefore examine whether these behavioral variables are significant predictors of religiosity.

Data description:

ATTNDPERRLS measures frequency of attendance at religious services (1 = more than once a week, 6 = never).

PRAY measures frequency of prayer outside services (1 = several times a day, 7 = never).

PRAC_B measures frequency of reading scripture (1 = at least once a week, 5 = never).

PRAC_D measures frequency of sharing religious views with people of other faiths (1 = at least once a week, 5 = never).

We test whether behavioral factors (attendance, prayer frequency, and religious practices) significantly predict religiosity.

Hypothesis Test: H_0: That none of these behaviors are associated with religiosity H_a: That at least one behavioral variable significantly predicts religiosity.

The results from the full model indicate that attendance (ATTNDPERRLS), prayer frequency (PRAY), and scripture reading (PRAC_B) are statistically significant predictors of religiosity ($p < 0.05$). Given the coding of religiosity (higher values indicate lower religiosity), these results suggest that more frequent religious practices are associated with higher levels of religiosity (i.e., more religious behavior). In contrast, PRAC_D is not statistically significant, suggesting that sharing religious views with others does not have a measurable relationship with religiosity in this model.

The model explains approximately 58.56% ($R^2 = 0.5856$) of the variation in religiosity, indicating a relatively strong relationship between behavioral practices and religiosity.

The results from the Protestant model indicate that attendance (ATTNDPERRLS), prayer frequency (PRAY), scripture reading (PRAC_B), and sharing religious views (PRAC_D) are statistically significant predictors of religiosity ($p < 0.05$). Given the coding of religiosity, where higher values indicate lower levels of religiosity, these results suggest that more frequent religious practices are associated with lower religiosity scores, indicating higher levels of religious engagement.

The model explains approximately 27.34% ($R^2 = 0.2734$) of the variation in religiosity, suggesting a moderate relationship between behavioral practices and religiosity within the Protestant subsample.

The results from the Catholic model indicate that attendance (ATTNDPERRLS), prayer frequency (PRAY), scripture reading (PRAC_B), and sharing religious views (PRAC_D) are statistically significant predictors of religiosity ($p < 0.05$). Given the coding of religiosity, where higher values indicate lower levels of religiosity, these results suggest that more frequent religious practices are associated with lower religiosity scores, indicating higher levels of religious engagement.

The model explains approximately 47.64% ($R^2 = 0.4764$) of the variation in religiosity, suggesting a relatively strong relationship between behavioral practices and religiosity within the Catholic subsample.

The results from the Muslim model indicate that attendance (ATTNDPERRLS), prayer frequency (PRAY), and scripture reading (PRAC_B) are statistically significant predictors of religiosity ($p < 0.05$). Given the coding of religiosity, where higher values indicate lower levels of religiosity, these results suggest that more frequent religious practices are associated with lower religiosity scores, indicating higher levels of religious engagement. In contrast, PRAC_D is not statistically significant, suggesting that sharing religious views with others does not have a measurable relationship with religiosity in this model.

The model explains approximately 40.63% ($R^2 = 0.4063$) of the variation in religiosity, suggesting a moderate to relatively strong relationship between behavioral practices and religiosity within the Muslim subsample.

The results from the Unaffiliated model indicate that some behavioral variables are statistically significant predictors of religiosity ($p < 0.05$). Given the coding of religiosity, where higher values indicate lower levels of religiosity, these results suggest that more frequent religious practices are associated with lower religiosity scores, indicating higher levels of religious engagement.

The model explains approximately 38.84% ($R^2 = 0.3884$) of the variation in religiosity, suggesting a moderate relationship between behavioral practices and religiosity within the Unaffiliated subsample.

Section 4: Political Predictors of Religiosity

It would appear that removal of the 99s which represent those who did not answer made a substantial difference on the fit of the model. Furthermore, support for gay marriage predicts religiosity well, as does support for transgender individuals. However, acceptance of homosexuality is an incredibly poor predictor of religiosity in the body public.

Alternatively, support for abortion, homosexuality, and gay marriage predicts spirituality but support for transgender individuals does not predict spirituality.

While support for abortion, same sex marriage, and homosexuality is a statistically significant predictor of religiosity in Catholics and people who are unaffiliated, support for these causes is not statistically significant in predicting religiosity in Protestants. Interestingly, greater acceptance of transgender individuals cannot be used to predict religiosity in people who are unaffiliated.

Section 5: Comprehensive Model

Finally, we construct a comprehensive model using 27 variables that incorporates behavioral measures, belief variables, and religious affiliation to determine which factors are most important in predicting religiosity.

With an R^2 of 0.71, we have pretty good confidence that this model can predict religiosity in individuals. An R^2 of 0.71 tells us that 71% of variance can be explained by the model and the variables that we presented it with. The results for spirituality were quite a bit lower, however, with an R^2 of just 43%.

Check for Assumptions of Regression: 1. Linearity 2. Nearly normal residuals 3. Constant variability

Looking at the p-values for the religiosity model, there were quite a few variables that were not impactful in predicting religiosity. These included FRMREL, RELTRAD, CURREL, CHATTEND, MARITAL, BIRTHDECADE, HISP, RACECMB, E2, USGEN, REG, PARTY, and GENDER. When these variables were removed, the R^2 value decreased to .64.

Conclusions

Across both the full sample and some religious subsamples, behavioral factors—including attendance, prayer frequency, and scripture reading—consistently emerge as significant predictors of religiosity. These findings suggest that religiosity is strongly shaped by observable religious practices, rather than being solely an internal belief system.

However, the strength of these relationships varies across religious traditions. The Catholic and Muslim subsamples show relatively stronger explanatory power, while the Protestant model shows a weaker relationship between behaviors and religiosity. The Unaffiliated group also demonstrates that behavioral practices remain relevant even in the absence of formal religious affiliation.

Overall, the results indicate that religiosity is best understood as a behaviorally structured phenomenon, but the degree to which behavior predicts religiosity differs across religious traditions. This suggests that religious tradition plays a moderating role in shaping how behaviors translate into levels of religiosity.

Similarly, depending on the religious tradition, certain political beliefs seemed to show a moderate correlation with religiosity and spirituality.

For the full models, religiosity was much easier to predict based on the behavioral, political and religious beliefs and factors than spirituality. One reason that this may be the case is potentially due to religious affiliation (with higher religiosity being already more rare) being associated with a specific set of beliefs that one subscribes to. On the other hand, one can claim to be spiritual to a certain extent, but this term is more vague in what it describes. Many classify themselves as "spiritual but not religious," a category that denotes vague spiritual feelings without adherence to a specific religion. It is important to note that the analogous "religious but not spiritual" designation is almost far less common than its counterpart and mostly nonsensical.